



"It's Never Lupus"

Ginger-derived compound demonstrates protective role against autoimmune disorders like Lupus in mice. <http://dgit.in/feb21-11>



Lowering The Efficacy

Depression and stress can alter the body's ability to develop an immune response to COVID-19 vaccines. <http://dgit.in/feb21-12>

nanoparticle. The nanoparticle is a clumped "cage" of 60 identical proteins, each of which acts like a tiny piece of velcro. Fragments of the spike proteins of various coronaviruses were taken, and were provided with a protein tag, which is like the other half of the velcro. These fragments are specifically the portions of the spike proteins that interact and bind with the cells, and are receptor binding domains (RBDs). The researchers created a pile of these modified RBD fragments. Crudely speaking, the researchers then take the nanoparticle cage and shake it all about in the portions of modified spike proteins. What results is a ball of pure horror, a nanoparticle core around which are stuck the spike proteins of several different coronaviruses. To test the approach, the researchers introduced such nanoparticles into mice. The approach worked, and the immune systems of the rats were able to fend off infections from the actual viruses after being treated by the vaccine. The researchers though, found something even better. The immune systems of the mice were able to fend off attacks from coronaviruses whose spike proteins had not even been put into the mix. This meant that the immune systems of the lab rats were trained on the common features of coronaviruses, and could potentially react to even novel strains that emerge.

MODELLING THE VIRUS

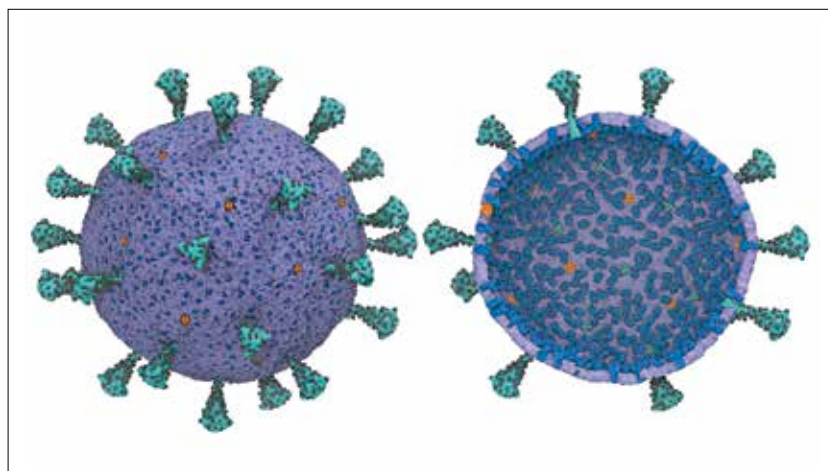
Researchers from the University of Chicago have modelled the entire

SARS-CoV-2 virus, and have made it available to researchers around the world. The model simulates the entire life cycle of the virus, including its interaction with hosts. To make a simplified version that is comprehensive but still possible to run on limited computational resources, the researchers used prior experience in a technique known as coarse graining. The most important characteristics of each component of the virus is identified, while the less important information is dropped. For example, modelling each individual atom in the virus as it interacts with the host's body would be prohibitively expensive in terms of computational power, as it would mean simulating billions of atoms. The researchers used previous experience on working with the HIV virus to develop the model. The model can be useful for investigating new drug

designs, as well as how new strains of the virus can emerge and spread. The model also has a framework that allows researchers to add in new information about the virus as and when the discoveries are made. The model is available on GitHub (<http://dgit.in/covsim>) for anyone to use.

ZOMBIE MOVIES AND THE VIRUS

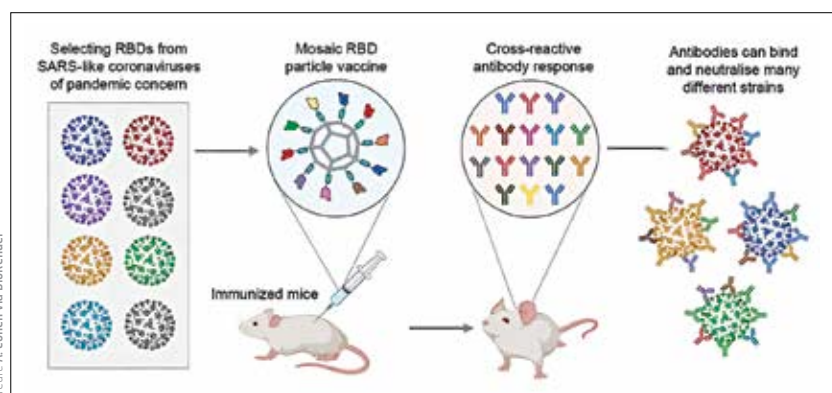
A team of researchers from Penn State wanted to investigate the factors that influenced the preparedness of the general public to a pandemic, that had nothing to do with their individual personalities. After conducting a study, they found something interesting. Those who were into watching post apocalyptic movies with aliens, zombies or viruses, were actually better prepared to handle the pandemic. Apparently watching these movies mentally prepares you to deal with the real thing. The researchers found that the more people had watched these movies, the better prepared they were during the pandemic. So does that mean those who are finding themselves suddenly at sea should fire up Netflix and grab a giant tub of popcorn? Not really. According to the researchers, watching the movies now cannot have a noticeable effect immediately. However, as the old saying goes, better late than never. At least the binge can prepare one for future challenges. 



The computational model of the SARS-CoV-2 virus

Credit: Yu et al., "A Multiscale Coarse-Grained Model of the SARS-CoV-2 Virus," Biophysical Journal (2021)

Credit: A. Cohen via BioRender



Schematic of the Caltech nanoparticle vaccine